

Solution Problem 4.1 (see also m-file)

Maximum of likelihood function

$$L(\hat{\mu}_{ML}, \hat{\Sigma}_{ML}) = \frac{1}{2\pi^{np/2}} e^{-np/2} \frac{1}{|\hat{\Sigma}_{ML}|^{n/2}}$$

Replacing $\hat{\Sigma}_{ML}$ with $S = \frac{1}{n-1} \sum (x_i - \bar{x})(x_i - \bar{x})^T$
and n with $n-1$ gives

$$L(\hat{\mu}, S) = \frac{1}{2\pi^{(n-1)p/2}} e^{-(n-1)p/2} \frac{1}{|S|^{n-1/2}}$$

Applied to the present two-sample case
the likelihood ratio becomes

$$\begin{aligned}\Lambda &= \frac{L(\hat{\mu}_1, \hat{\mu}_2, S_{pooled})}{L(\hat{\mu}_1, S_1) \cdot L(\hat{\mu}_2, S_2)} \quad \leftarrow H_0 \text{ true (restricted max)} \\ &= \frac{(2\pi)^{(n_1-1)p/2} (2\pi)^{(n_2-1)p/2} e^{-(n_1+n_2-2)p/2}}{(2\pi)^{(n_1+n_2-2)p/2} e^{-(n_1-1)p/2} e^{-(n_2-1)p/2} \cdot \frac{|S_1|^{n_1-1/2} |S_2|^{n_2-1/2}}{|S_{pooled}|^{n_1+n_2-2/2}}} \quad \leftarrow \text{general (unrestricted max) (independence)} \\ &= \frac{|S_1|^{n_1-1/2} |S_2|^{n_2-1/2}}{|S_{pooled}|^{n_1+n_2-2/2}}\end{aligned}$$

$$-2 \log \Lambda = (n_1+n_2-2) \log |S_{pooled}| - (n_1-1) \log |S_1| - (n_2-1) \log |S_2|$$

QED.

Since $df_{denom} - df_{nom} = df_{\Sigma} = \frac{1}{2} p(p+1)$
which is the number of unique entries in $p \times p \Sigma$
the test becomes s (except for correction factor r)

$$\begin{aligned}\text{Reject } H_0: \Sigma_1 = \Sigma_2 & \text{ if} \\ -2 \log \Lambda & > \chi^2_{\frac{1}{2}p(p+1)}(\alpha)\end{aligned}$$

(See m-file for numerical example)